Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/06/2008
PI Name:	Li, Ben Q. Ph.D.		
Project Title:	Study of Magnetic Field Effects on Convection and Solidification in Normal and Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	benqli@engin.umd.umich.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	313 593 5465
Organization Name:	University of Michigan-Dearborn		
PI Address 1:	Dbn-Col Engr & Comp Sci		
PI Address 2:	1344 EC 1491		
PI Web Page:			
City:	Dearborn	**State:**	MI
Zip Code:	48128	**Congressional District:**	15
Comments:	Moved to University of Michigan, per D. Gillies/MSFC (4/07), from Washington State University.		
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Banks, Curtis	**Contact Phone:**	256 544-4437
Contact Email:	curtis.e.banks@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	de Groth, H C. (NASA Glenn Research Center)		
Grant/Contract No.:	NAG8-1693		
Performance Goal No.:			
Performance Goal Text:			

The theoretical part of the proposed research will involve developing mathematical models to represent the magnetic damping effects on the convective flow, temperature distribution, and solute profiles in normal and micro gravity, and to conduct extensive numerical simulations to quantify the damping and/or enhancement effects. The work includes the following components: (i) carrying out additional numerical simulations to characterize the 3-D melt flow structure in the physical model driven by oscillating temperature gradients, (ii) carrying out additional simulations, using the existing 2-D solidification model, of solidification phenomena as affected by oscillating thermally induced flows, (iii) developing a 3-D numerical solidification model for oscillating flow affected solidification phenomena and comparing predictions with laboratory measurements, and (iv) developing numerical models for linear analysis of melt flows in the presence of magnetic field in finite geometry and for fully nonlinear analysis to quantify the convective flow enhancements by an applied DC magnetic field ?the linear analysis provides the starting conditions for the nonlinear

Task Description:	analysis. In parallel to the above theoretical studies, ground-based laboratory experiments will be conducted. These measurements will be made in transparent fluids, and in mercury pools with and without the presence of a magnetic field. These measurements are intended to verify the predictions from the above analyses and also to provide a database for future model development or models developed by other research groups. As the basic experimental systems and measuring infrastructure are already in place, only some modifications are needed to achieve the required objectives. Oscillating convective flows in transparent fluids will be measured using the PIV system and oscillating velocity field and damping effects on the flow in mercury will be measured using hot film velocimeters. These measurements will be used to verify the 2-D and 3-D models of magnetic damping for both steady state and transient flows associated with g-jitter and to help understand and quantify the transport phenomena in melt processing systems. The objectives of the proposed research are to: (1) conduct measurements of 3-D oscillating melt flows in physical models, and to verify and refine the 3-D melt flow model developed for g-jitter induced flows against the measured data; (2) assess the effect of oscillating convective flows on solidification in a microgravity environment through numerical modeling and ground-based laboratory measurements; and (3) understand and quantify the effect of convective flow enhancement in a DC magnetic field, whose main action has so far been understood to provide damping effects on convection. The proposed research will generate information that is critical for both fundamental understanding and quantitative assessment of magnetic field effects on oscillating and/or random fluid motions induced by g-jitter encountered in microgravity melt processing systems. The numerical models and experimental setups may be used to further explore new concepts and ideas for design and development of magnetic damping facilities for space and terrestrial applications, and for a rational planning of microgravity and ground-based solidification experiments and interpreting the measured results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No final report received, but publications resulting from this NASA-funded research are listed in the Bibliography below.
Bibliography Type:	Description: (Last Updated: 02/07/2014)
Articles in Peer-reviewed Journals	Li K, Li BQ, Handa J, de Groh HC. "Three-dimensional numerical simulation of g." International Journal of Numerical Methods for Heat & Fluid Flow 2005;15(8):872-93.http://dx.doi.org/10.1108/09615530510625138 , Aug-2005
Articles in Peer-reviewed Journals	Shu Y, Li BQ, Ramaprian BR. "Convection in modulated thermal gradients and gravity: experimental measurements and numerical simulations." International Journal of Heat and Mass Transfer, 2005 Jan;48(1):145-60. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2004.08.010 , Jan-2005
Articles in Peer-reviewed Journals	Shu Y, Li BQ, de Groh HC. "Convection and solidification in constant and oscillating thermal gradients: Measurements and smulations." Journal of Thermophysics and Heat Transfer 2005 Apr-Jun;19(2):199-208. , Apr-2005